Griffin Garland

New York, NY | griffingarland.com | linkedin.com/in/griffingarland

Summary

Software engineer with 10+ years of experience building C++ and Python systems in financial technology, currently leading the static analysis and automated refactoring team at Bloomberg. Depth in static analysis, large-scale migration, and high-performance distributed systems.

Technical Skills

Languages: C++, Python, C, SQL, Bash
Systems: Linux, Distributed Systems, Redis, RabbitMQ, Kafka, RESTful APIs, AWS
Build & Tooling: Static Analysis, CMake, Git, SVN, CI/CD, Docker, Claude Code

Experience

Software Engineering Team Lead — April 2024 – Present
Bloomberg LP (Developer Experience Department) — *New York City, NY, USA*

- Drive architecture and technical direction for static analysis across Bloomberg's codebase as team lead of STAAR (Static Analysis and Automated Refactoring); leading 6 engineers plus 2 contractors.
- Operate and scale the STAAR pipeline, a distributed static analysis system running over 2.5 million jobs per week across tens of thousands of independent repositories, exposing issues via Bloomberg's initiative reporting dashboard.
- Integrate static analysis into Bloomberg's meta-build system (berg) and CI platform (Devise), creating a unified interface for running and getting static analysis results across all supported languages.
- Delivered coverage for all 6 of Bloomberg's tier one and tier two languages to satisfy EU DORA and SSDF regulatory requirements.
- Partnered with the CISO to detect over 500 vulnerability/language permutations, including SQL injection and memory safety issues.

Software Engineering Team Lead — May 2022 – March 2024
Bloomberg LP (Trading Solutions Department) — *New York City, NY, USA*

- Owned technical direction and delivery for the Options and Loans team within AIM, Bloomberg's buy-side OMS; leading 5 engineers.
- Co-owned the C++ "Order Trade Entry" ticketing engine processing up to hundreds of millions of transactions daily.
- Led delivery of Bond Forwards and Loans support in the C++ ticketing engine, a gating requirement for institutional clients to onboard onto AIM; drove design and execution while implementing core order/trade persistence logic hands-on.
- Partnered with AIM product owners to scope and prioritize the Options and Loans product roadmap, converting client requirements into technical designs and delivery plans.

Senior Software Engineer — October 2017 – May 2022
Bloomberg LP (Trading Solutions Department) — *New York City, NY, USA*

- Tech lead for the cross-department integration between BSKT (Bloomberg's basket-trading product) and AIM, eliminating manual re-entry and bringing BSKT clients onto AIM.
- Implemented a RabbitMQ-based queueing system for AIM's C++ ticketing engine, allowing durable messaging with high scaling.
- Designed and implemented a Redis-backed distributed caching layer for the "Order Defaults" settings service, reducing settings page load time by a factor of 5 and latency for specific cached request types by a factor of 50, with a cache hit rate of over 99%.
- Migrated 21 core services and libraries to Linux as part of Bloomberg's company-wide migration off legacy UNIX OSes (Solaris and AIX).
- Led the migration of JavaScript UI components to a new WebKit-based rendering engine as part of a company-wide initiative.
- Selected as a technical leader of the Trading Solutions refactoring working group, helping developers migrate their legacy C++ to Linux.

Cloud Integration Software Engineer — April 2017 – August 2017
Solace Corporation — *Kanata, ON, Canada*

- Implemented LDAP authentication in Java for the Solace Pivotal Cloud Foundry (PCF) tile.
- Automated test environment provisioning with AWS CloudFormation, standing up pre-configured EC2 instances on demand.

Software Developer — June 2015 – March 2017
Solace Corporation — *Kanata, ON, Canada*

- Maintained and extended the C++ configuration management software for the Solace Messaging Appliance.
- Designed and implemented a RESTful configuration management API in Python.
- Redesigned the auto-generated Python command handling infrastructure around a polymorphic approach.

Education

University of Toronto — September 2010 – June 2015
Bachelor of Applied Science, Electrical and Computer Engineering — *Toronto, ON, Canada*

- 16-month co-op between 3rd and 4th year at Algorithmics (an IBM company).